

MANAS GANESH DASARI

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OVERVIEW

Computer Science student focused on software engineering, AI/ML, systems, and optimisation. Builds technically substantial applications spanning real-time collaboration, stochastic modelling, and data-driven optimisation. Actively builds open-source software and contributes to community-driven projects, with a strong interest in practical engineering and systems development.

EDUCATION

Amrita School of Engineering, Bangalore: B.Tech. Computer Science | Expected 2028

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Architecture, Machine Learning, Neural Networks & Deep Learning, Probability & Statistics, Computer Networks

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL, Bash/Shell, Go, Visual Basic

AI / ML: NumPy, Pandas, scikit-learn, Jupyter Notebook, [Kaggle](#), [Hugging Face](#)

Web: React, Vite, Node.js, Express, Astro, FastAPI, HTML5, CSS3, REST APIs

Systems: Linux, Docker, Git, QEMU, UTM

TECHNICAL PROJECTS

[Sprout](#): **Codebase Intelligence CLI: Go, Git, Dependency Graphs, MCP**

- Built an open-source codebase mapping CLI that combines git-aware repository trees, dependency analysis, PR visualisation, and AI-oriented codebase context.
- Engineered a deterministic file-level dependency graph with forward/reverse adjacency and ranked entry-point analysis for repository navigation.
- Developed `--entry`, `--diff`, and MCP workflows for repository orientation, change analysis, and compact agent context without dumping source bodies.

[Flow](#): **Decentralised Real-Time Virtual Study Platform:** JavaScript, Node.js, Express, Supabase, WebRTC, Groq Llama, PostgreSQL

- Architected a peer-to-peer study platform using WebRTC and Supabase Realtime for synchronised collaboration.
- Implemented collaborative Pomodoro timers, task management, AI-assisted chat, and study rooms backed by PostgreSQL.
- Added secure authentication and TURN-assisted WebRTC connectivity with a responsive collaborative interface.

[Atmos](#): **Urban Air Quality Optimisation Platform:** Python, FastAPI, NumPy, Optimisation Modelling

- Built a system to simulate and optimise urban air-quality sensor placement using real-time WAQI data and traffic-weighted zone analysis.
- Applied optimisation modelling to propose sensor networks designed to improve coverage while reducing cost relative to existing CAAQMS stations.

Dynamic Stochastic Quantitative Trading Engine: Pine Script, Statistical Modelling, LaTeX | [Patent under review](#)

- Developed a real-time Pine Script trading engine using KNN price-pattern recognition, Markov-chain volatility regime detection, and Poisson-based risk controls.
- Achieved a 1.875 Profit Factor with 0.44% maximum drawdown during simulation testing and built an interactive Kelly sizing dashboard.

OPEN SOURCE

-> [Exercism](#) | VB.NET Track Contributor

- Actively contribute to the Exercism VB.NET track through pull requests involving bug fixes, exercise test-suite updates, and track maintenance.
- Actively working on assigned issues across the VB.NET track, with 10+ issues addressed through fixes, test updates, and maintenance contributions.
- Contributions have been reviewed and merged into the VB.NET track, with work credited in the project's contributor records.

RESEARCH & COMMUNITY

- Research: Developing two technical manuscripts on urban air-quality sensor optimisation and stochastic financial forecasting.
- Volunteer, [Barket](#) | Jun 2026–Present: Support animal adoption events, volunteer logistics, and community outreach.